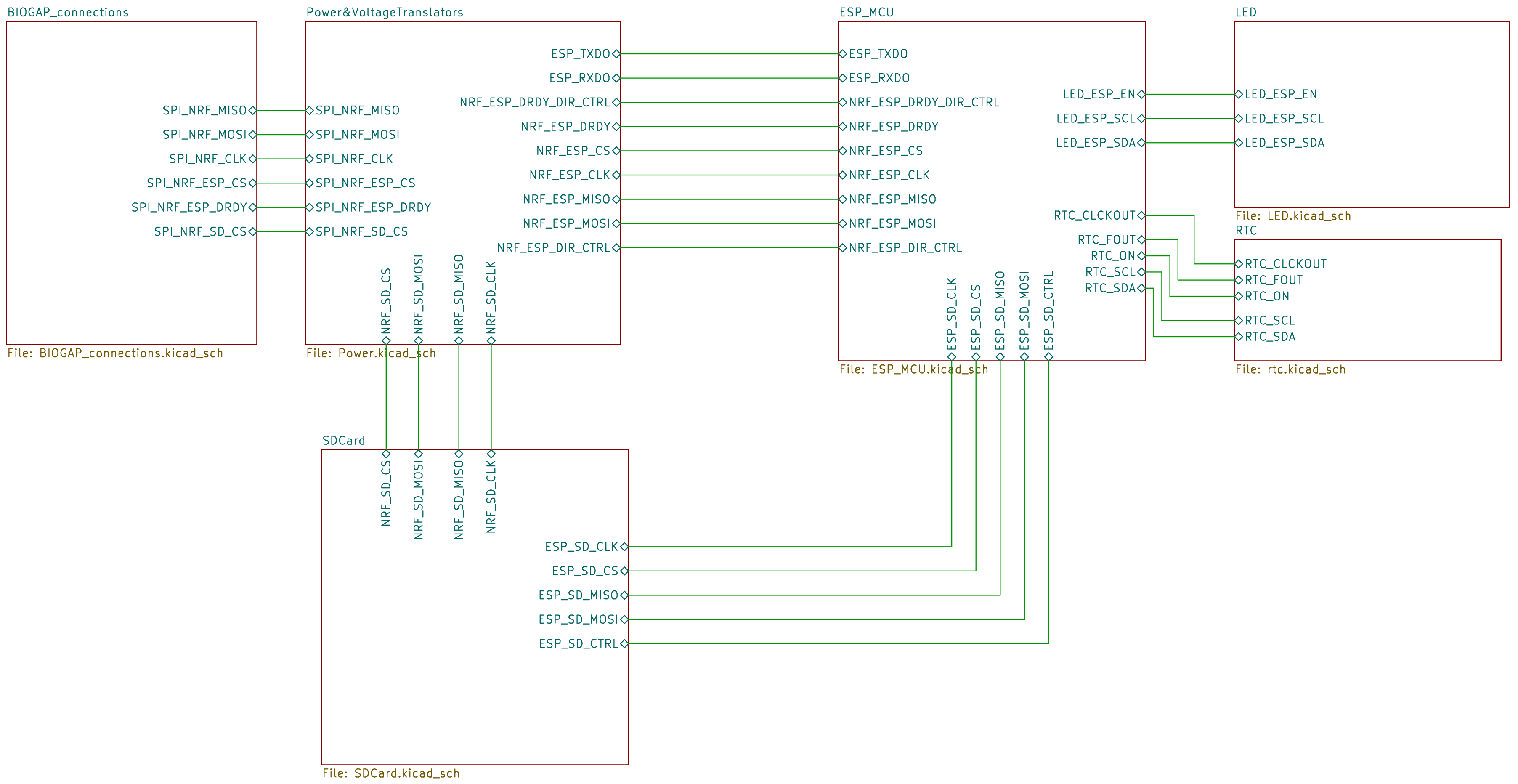


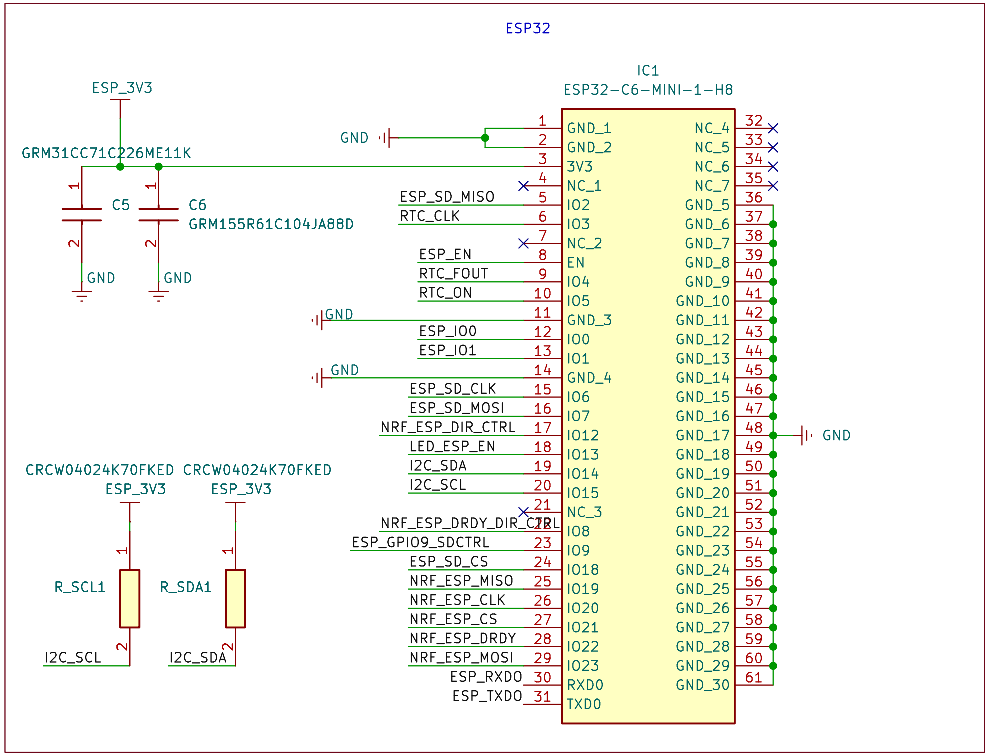
ETH Zurich

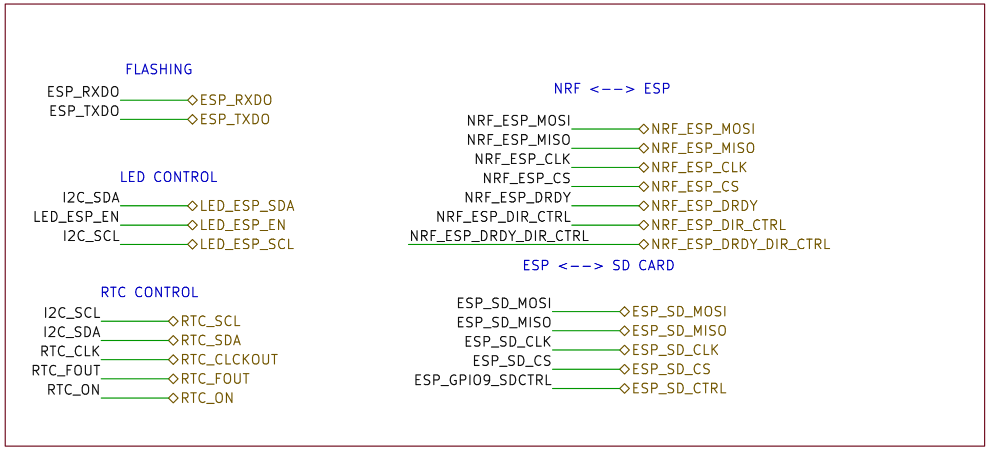
Sheet: /

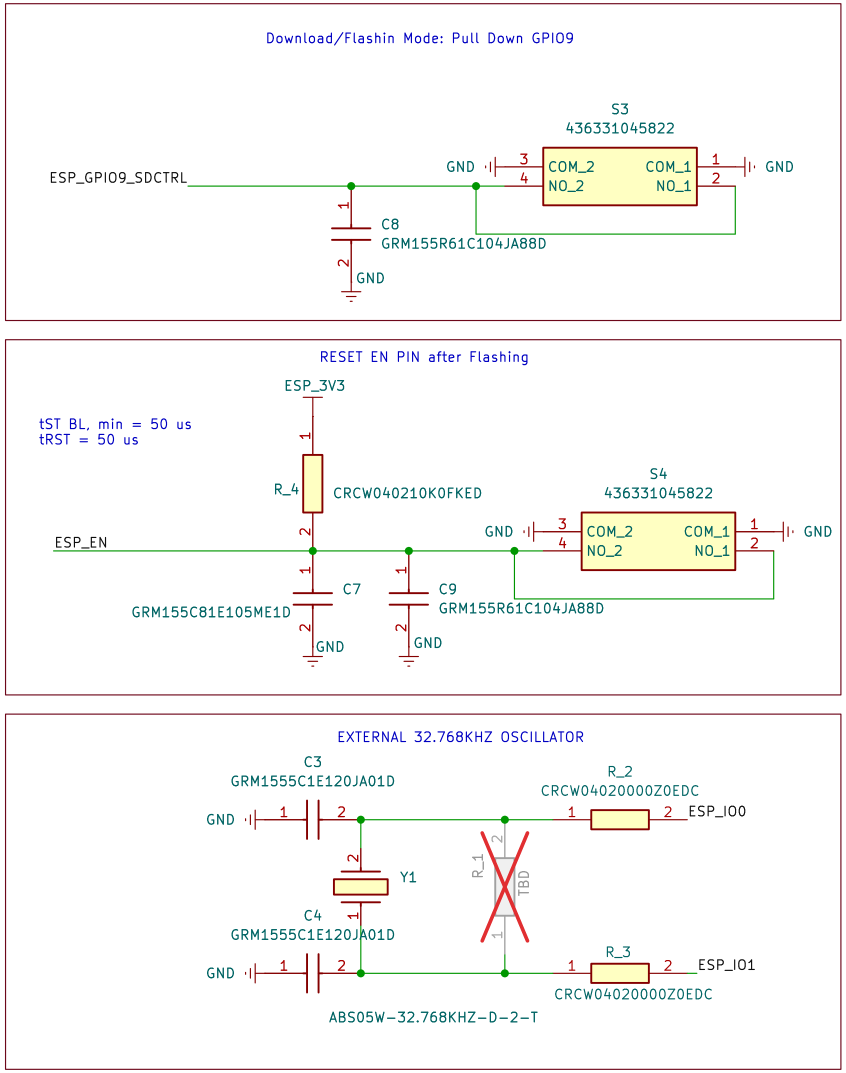
File: wifi_shield_v1.kicad_sch

Title: BIOGAP Ultra WiFi Shield

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.8		Id: 1/7





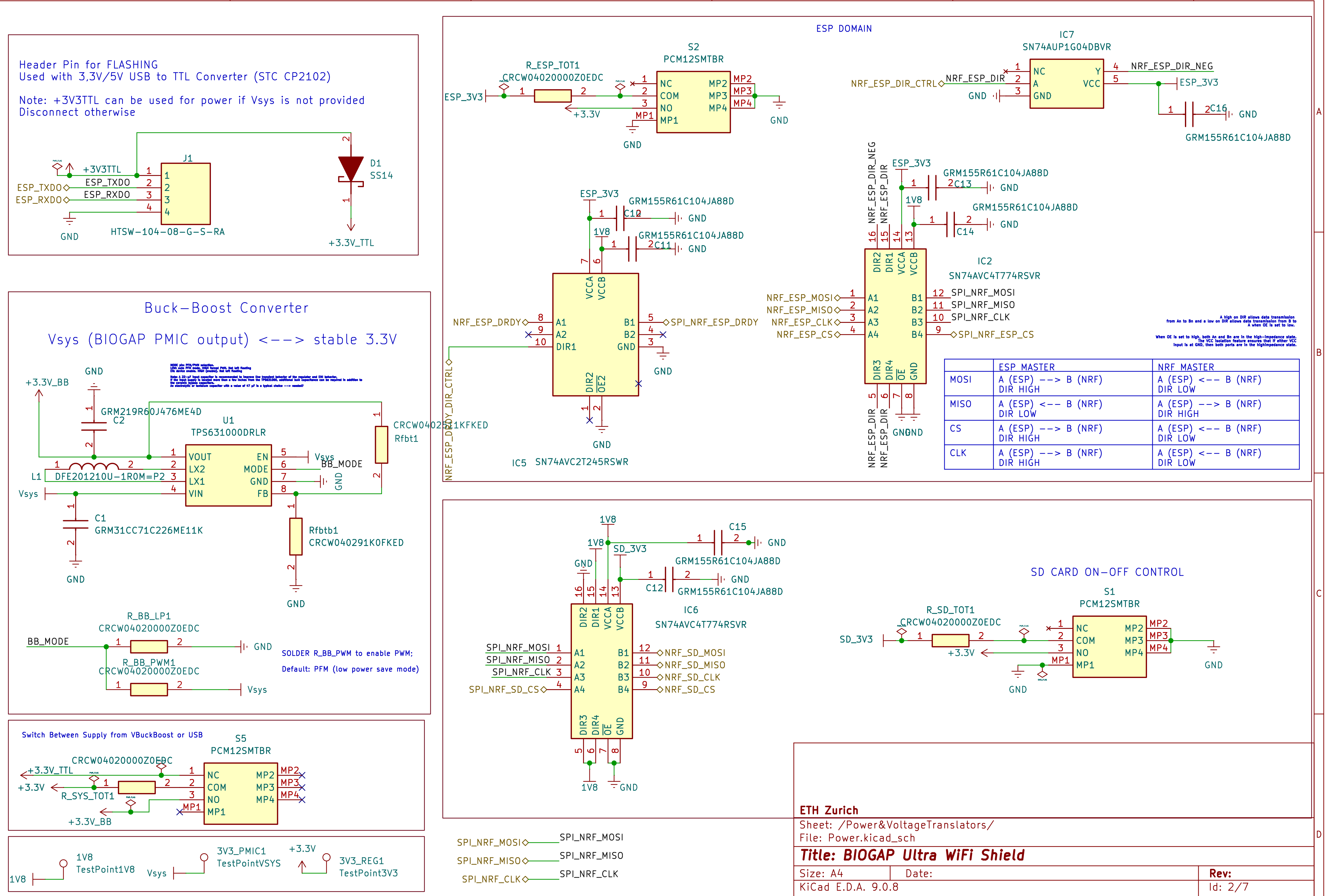


ETH Zurich

Sheet: /ESP_MCU/
File: ESP_MCU.kicad_sch

Title: BIOGAP Ultra WiFi Shield

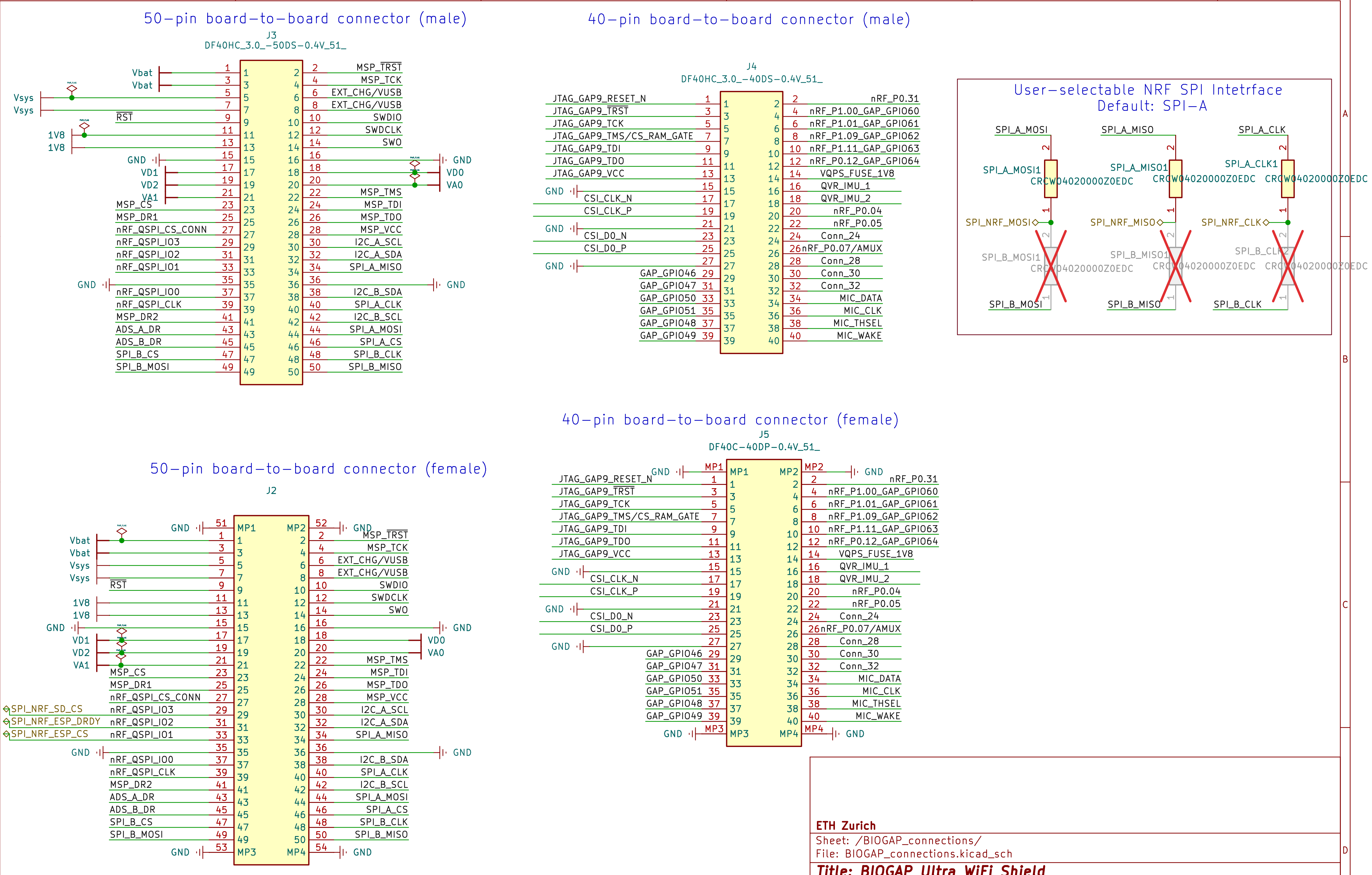
Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.8		Id: 1/7



	ESP MASTER	NRF MASTER
MOSI	A (ESP) --> B (NRF) DIR HIGH	A (ESP) <-- B (NRF) DIR LOW
MISO	A (ESP) <-- B (NRF) DIR LOW	A (ESP) --> B (NRF) DIR HIGH
CS	A (ESP) --> B (NRF) DIR HIGH	A (ESP) <-- B (NRF) DIR LOW
CLK	A (ESP) --> B (NRF) DIR HIGH	A (ESP) <-- B (NRF) DIR LOW

SPI_NRF_MOSI — SPI_NRF_MOSI
SPI_NRF_MISO — SPI_NRF_MISO
SPI_NRF_CLK — SPI_NRF_CLK

ETH Zurich
Sheet: /Power&VoltageTranslators/
File: Power.kicad_sch
Title: BIOGAP Ultra WiFi Shield
Size: A4 Date: Rev:
KiCad E.D.A. 9.0.8 Id: 2/7



ETH Zurich

Sheet: /BIOGAP_connections/

File: BIOGAP_connections.kicad_sch

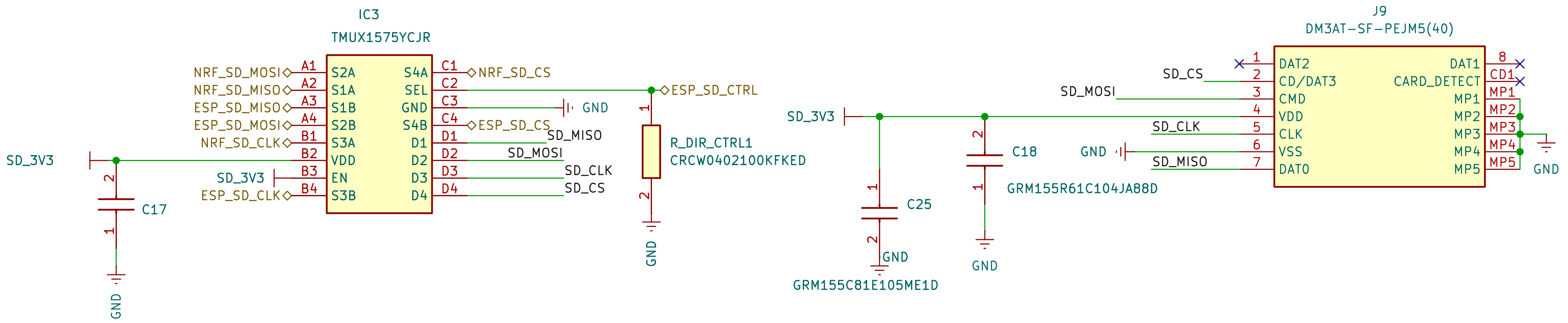
Title: BIOGAP Ultra WiFi Shield

Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.8		Id: 3/7

EN: Active high, internal 6MO pull down to GND
SEL: Internal 6MOhm pull-down to GND
Default: Pulled down, NRF writes to SD CARD
If ESP pin HIGH -> ESP writes to SD card.

SPI NAME	SD Pin Name	Direction	Function
CS	DAT3	MCU -> SD	Select Device
SCLK	CLK	MCU -> SD	Clock
MOSI	CMD	MCU -> SD	Write data
MISO	DAT0	SD -> MCU	Reads Data

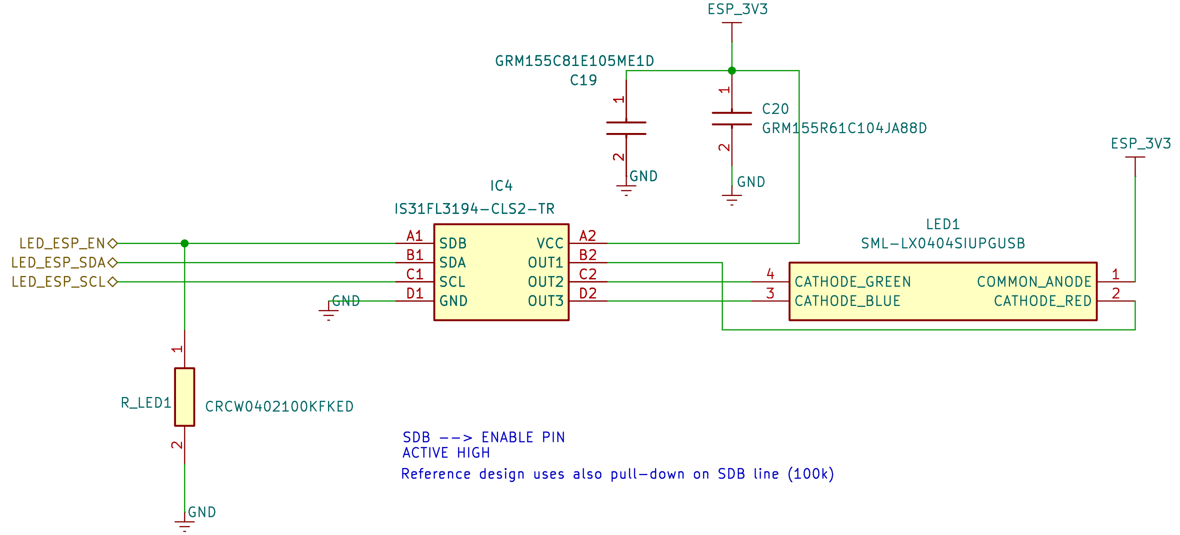
SD CARD INPUT SELECTION (NRF VS ESP)



ETH Zurich
Sheet: /SDCard/
File: SDCard.kicad_sch

Title: BIOGAP Ultra WiFi Shield

Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.8		Id: 5/7



ETH Zurich

Sheet: /LED/
File: LED.kicad_sch

Title: BIOGAP Ultra WiFi Shield

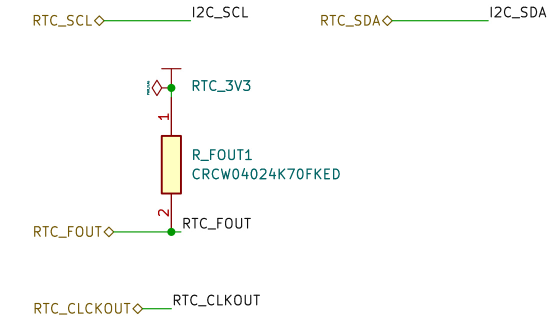
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.8		Id: 6/7

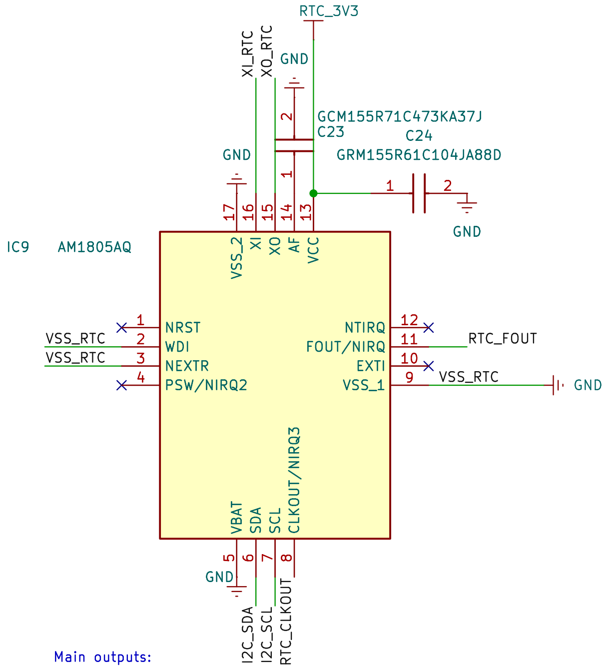
Page 42 datasheet
SDA and SCL lines must be connected to a positive supply voltage via a pull-up resistor.
I2C termination resistors should be above 2.2kOhm.

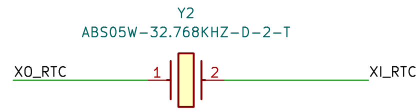
For systems with short I2C they can be as large as 22kOhm (400Khz operation) and 56kOhm (100kHz operation).
ESP has already an internal weak-pull up resisto (45kOhm)

R* for I2C lines.
Default: Not Mounted, ESP internal pull-up (56kOhm) enabled

400KHz operation: mount 22KOhm resistor

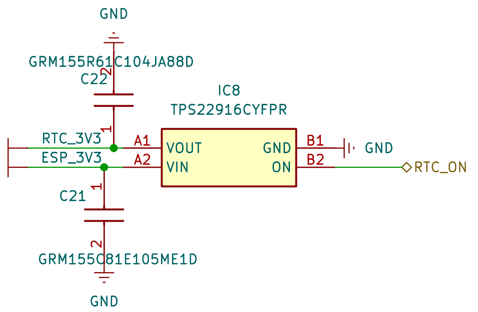






LOAD SWITCH

To actively enable/disable RTC from ESP



Main outputs:

FOUT/NIRQ: Int 1/function output. Can be used to send an interrupt to the ESP.
- Asserted low on power-up,high when device becomes accessible.
- We can use it as wake-up alarm for the ESP during deep-sleep
Open drain, needs pull-up when used.
nIRQ3/CLKOUT: Int 3 / clock output. Square Wave Output Connection.

Others:
nTIRQ: timer interrupt output connection; drivers the active low nTIRQ signal.
If used, an external pull-up must be added. Else, can be left floating

VBAT --> if a backup battery is not present, VBAT must be connectedd to VSS (gnd).
Note: can also be used to provide the analog input to the internal comparator (needed?)
nRST: External reset. If not used, left floating.
nEXTR: External reset input connection. If not used, connect to VCC or VSS
WDI: watchdog timer reset input. It must not be left floating; connect to either VCC or VSS if not used.

Application note: https://ambiq.com/wp-content/uploads/2020/08/Artasie-AMx8x5-Hardware-Design-Guidelines-App-Note.pdf

ETH Zurich
Sheet: /RTC/
File: rtc.kicad_sch
Title: BIOGAP Ultra WiFi Shield
Size: A4 Date: Rev:
KiCad E.D.A. 9.0.8 Id: 7/7